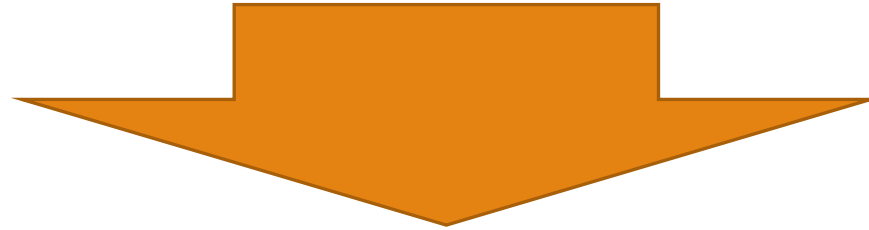


IMIC: Intelligent Manufacturing
International Contest - towards the
gradual design of a new benchmark

Motivations & Ambitions

Validation using real data from industrial case studies

Lack of a suitable, comprehensive, and standardized benchmark for real industrial cases in smart manufacturing



Creation of a benchmark for Intelligent Manufacturing

Representative enough to real-world applications

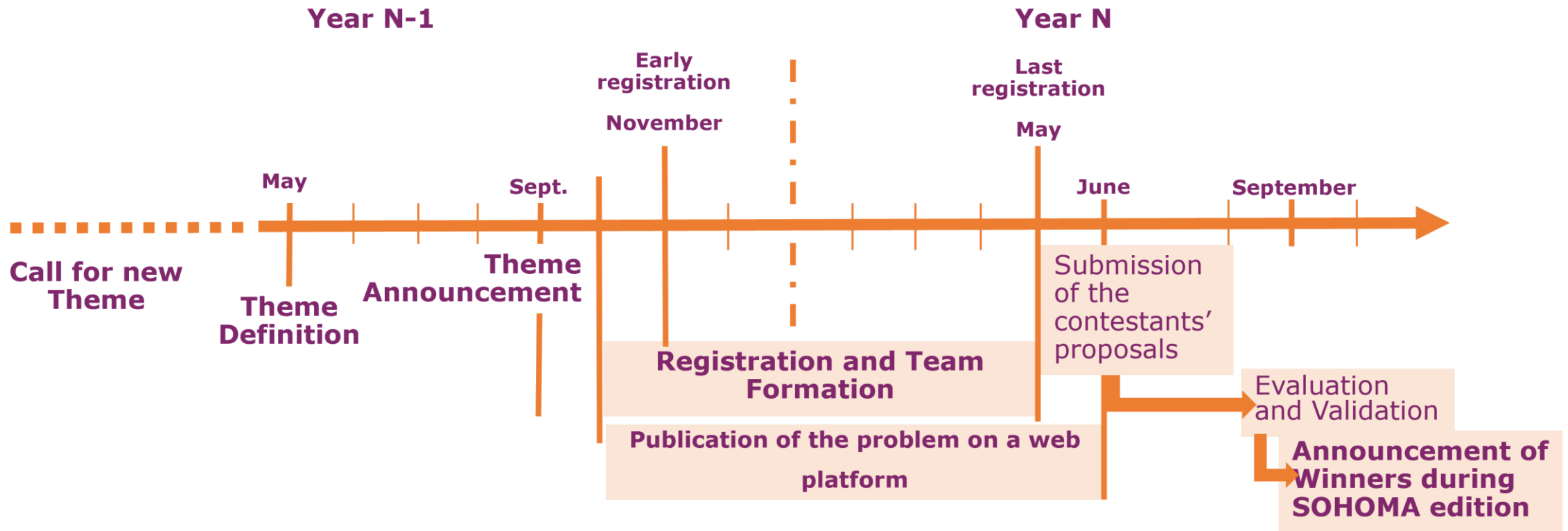
Relevant and adaptable to the dynamic nature of manufacturing technologies

Actually used by the community! Why not let you play with it?

We need you to build a benchmark



Phases of the contest



Theme of IMIC'25

Context

- *A cart loads trolleys with mail containers*
- *30 kg of bins per day*
- *250 destinations, only 10 around the robot (max 15 bins per trolley)*
- *Around 6 seconds to handle each box*

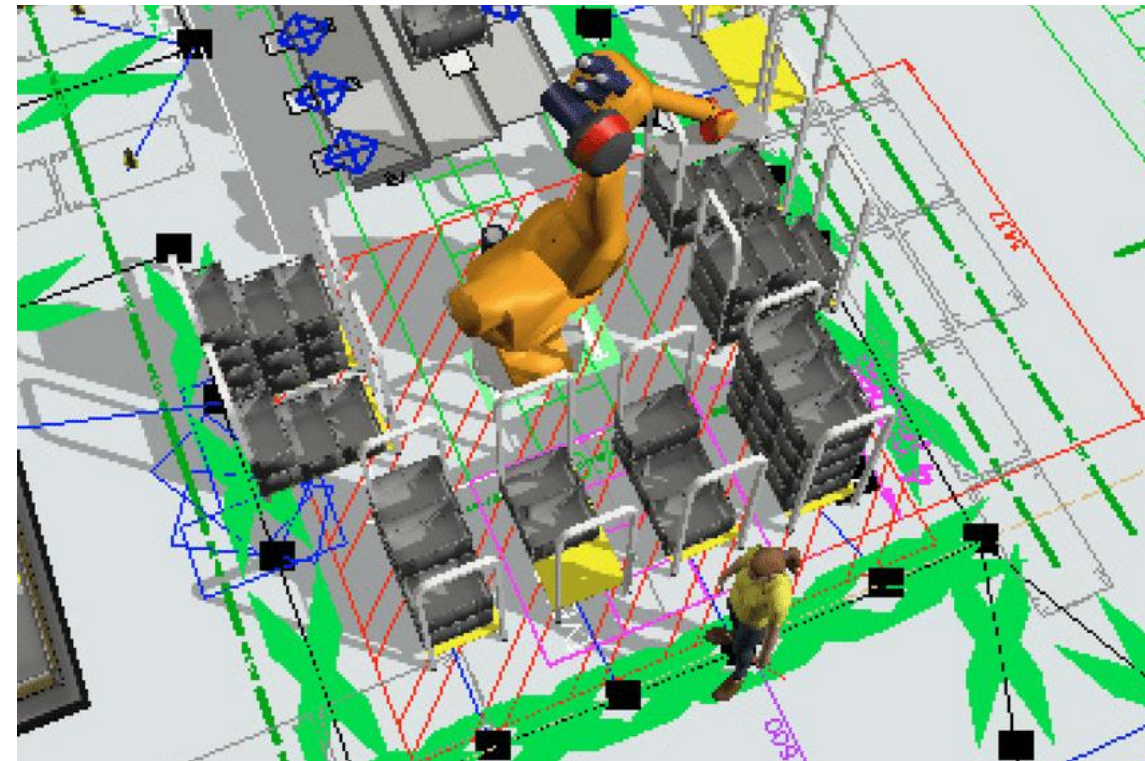
Decision to be made

- *Which trolleys around the robot?*
- *At what time ?*

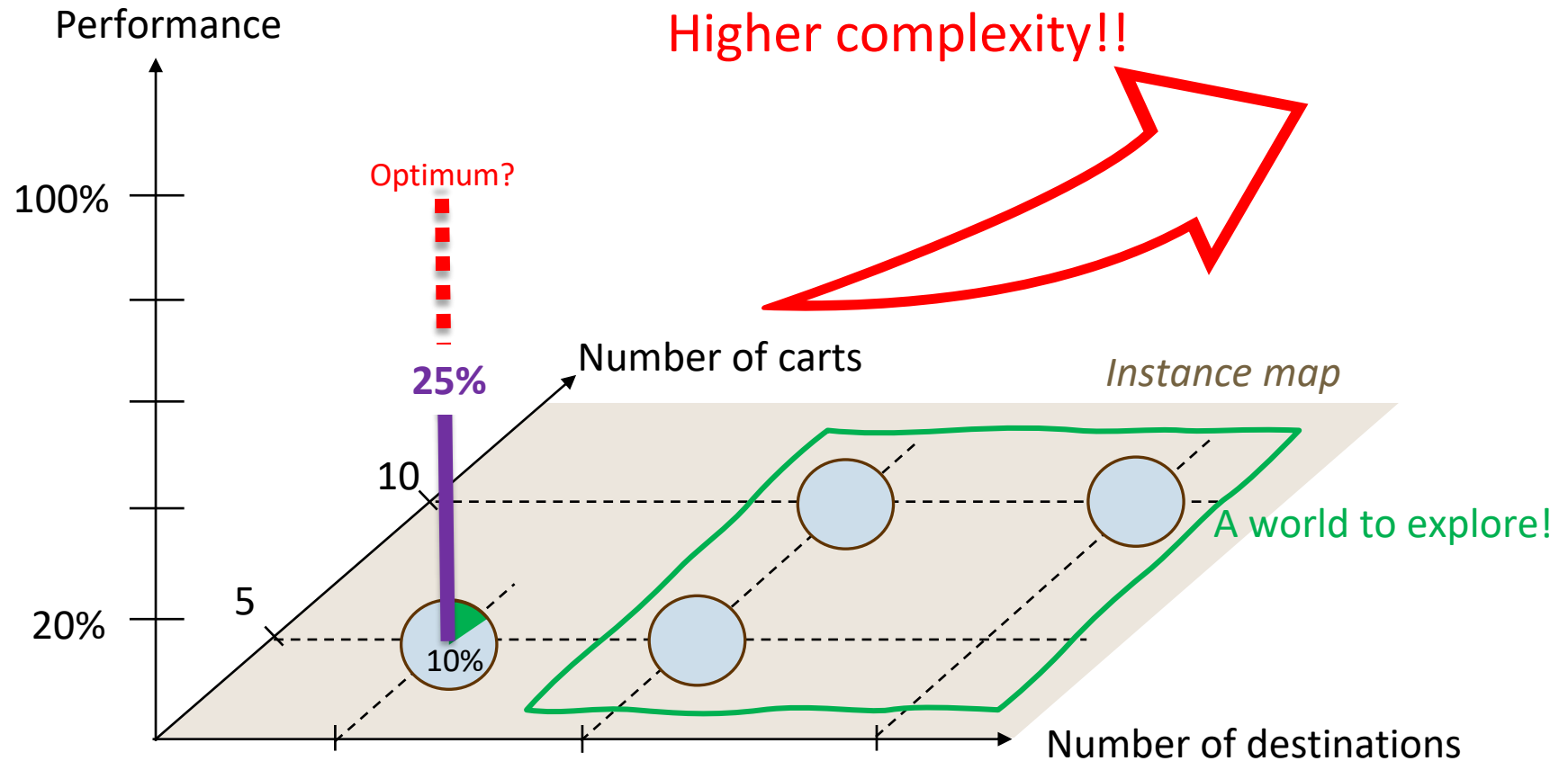
Data

- *3 real sorting days*
- *1 bin = (date of arrival, destination)*
- *Possibility of generating random days*

Flexibac (La Poste)



Current edition: IMIC26



What happened during 2025?

- Only 2 proposals solving one instance (10% of one case, **2.5%** total)
- Online but not **offline** proposals → What about the other instances?

Next edition: IMIC26



Next edition: Flexibac+!

Motivations for 2026:

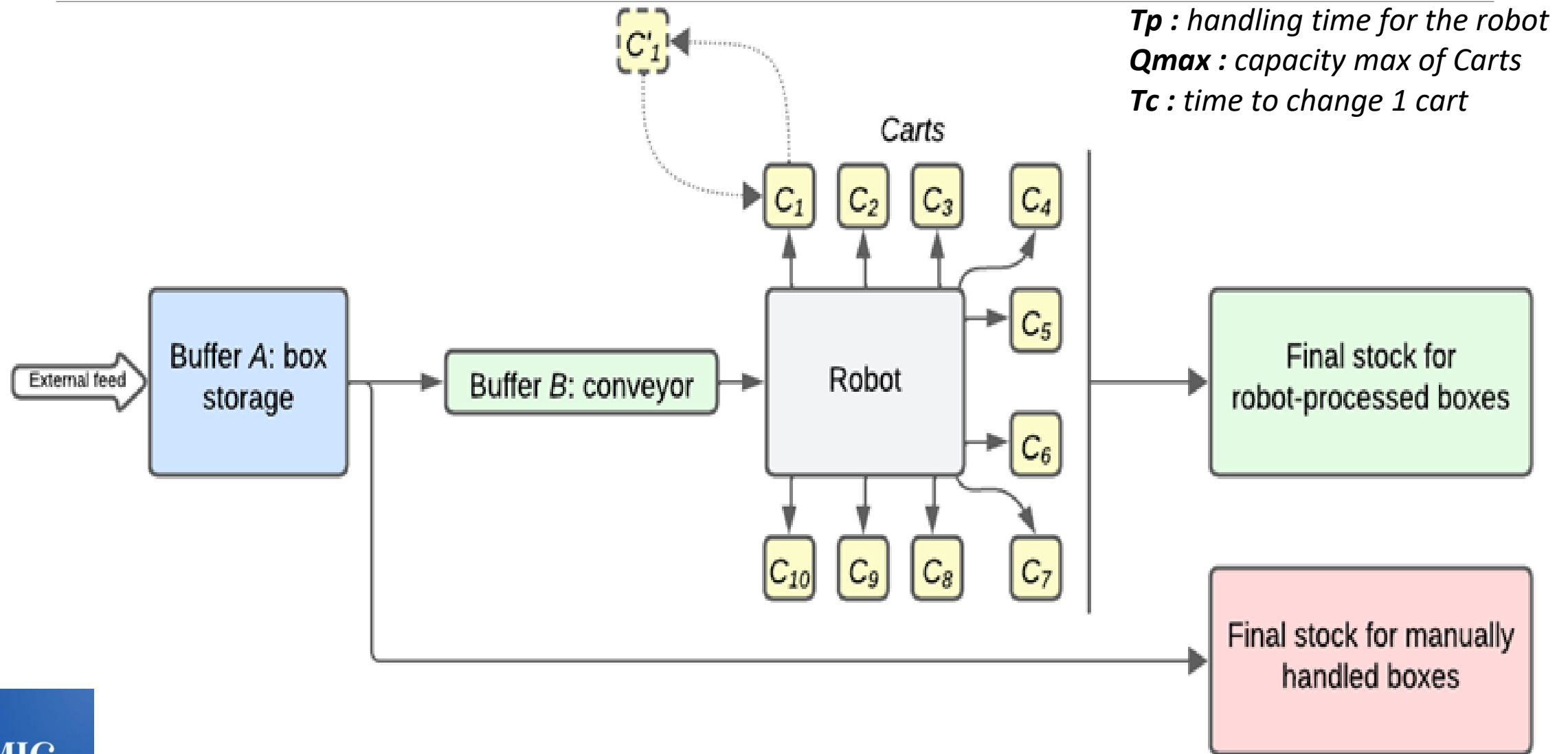
- Build the complete benchmark
- Explore the maximum set of instances
- Compare different approaches, online VS offline to prove optimality

Let's collaborate together !

- Regular meetings during the period
 - Kick-off meeting
 - « How-is-it-going? » meeting
 - Finalisation meeting
- Write a common publication:
 - all contestant with results will be part of a publication on the benchmark. *Vector to identify?*



Topology



Problem parameters

Inputs

- $A_{max}, B_{max}, T_p, Q_{max}, T_c$;
- A list of input dates and destinations of the boxes in buffer A;
- Some information about the distribution of destinations in the list.

MODEL
made by
the contestants

Outputs

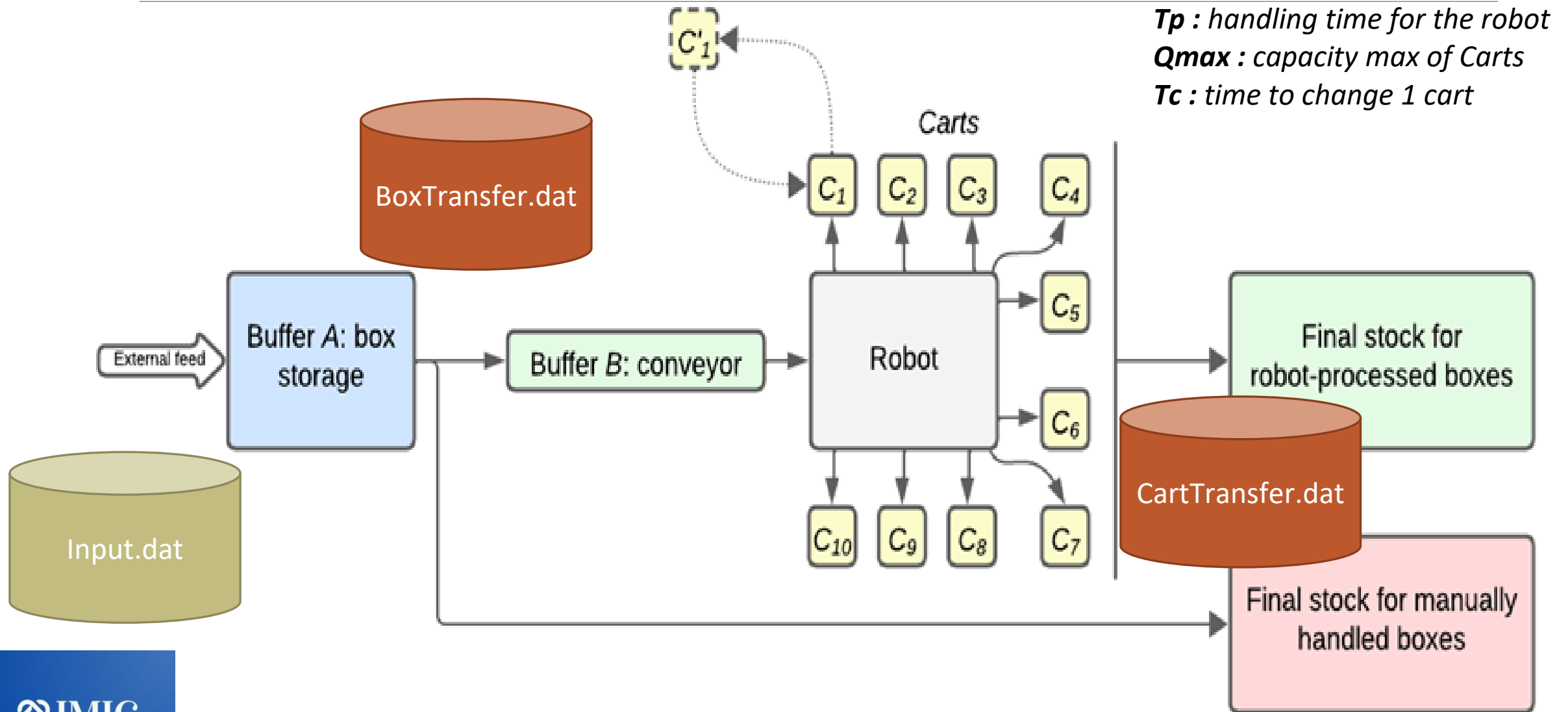
- List of dates of transfer of each box from buffer A to buffer B
- List of dates of cart changes at each position

Key Performance Indicators

1. The number of boxes the robot handles in a 24h period (max)
2. The number of carts that were brought to the robot (min)

Simulator

Topology



A_{max} : capacity max of A

B_{max} : capacity max of B

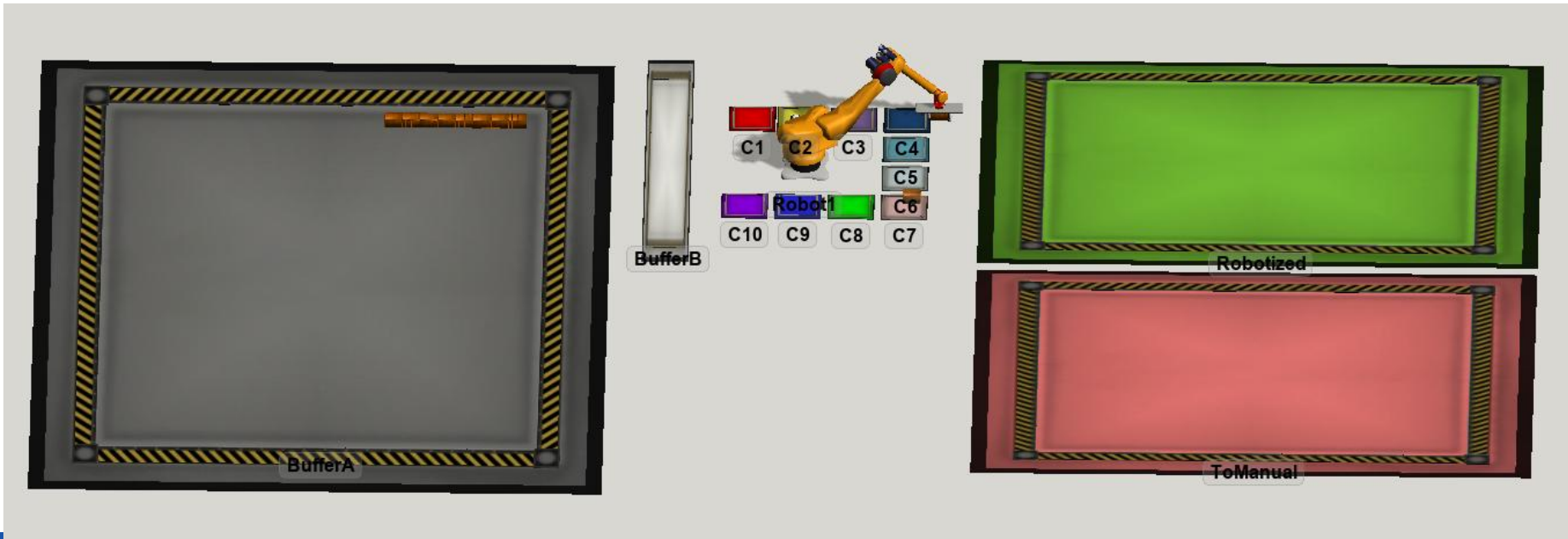
T_p : handling time for the robot

Q_{max} : capacity max of Carts

T_c : time to change 1 cart

Demonstration of the simulator

- Model developed in Flexsim
- Made available to the community, freely usable without licence



Dashboard

Containers successfully treated

Buffer	Final content
Robotized	40
ToManual	27

Containers unsuccessfully treated

Buffer	Final content	Maximum content
BufferA	5	11
BufferB	6	7
C1	4	4
C2	3	5
C3	2	5
C4	2	2
C5	0	6
C6	1	7
C7	1	6
C8	0	2
C9	4	4
C10	5	5

Number of cart changes

Name	Current
NumberOfChanges	10

Two categories for this year challenge (Online & Offline) [1/2]

- Different problem size (Small, medium and large instances) for each category
- Report the time taken by the proposed approaches to calculate the solution.

1. Online category

- A real industrial case study of a mail sorting system.
- Focuses on developing responsive approaches for a mail sorting system
 - Minimal worker intervention and maximal robot utilization.
- **No prior knowledge** of the list of input dates and destinations of boxes in buffer A.
 - A list of tuples that represents the arrival time and destination of a box in buffer A.
 - A distribution of destinations categorized into pareto classes based on frequency
- The solution should be responsive to the real-time arrival of boxes and their destinations, ensuring efficient and timely sorting
- Time constraint (24h)

Two categories for this year challenge (Online & Offline) [2/2]

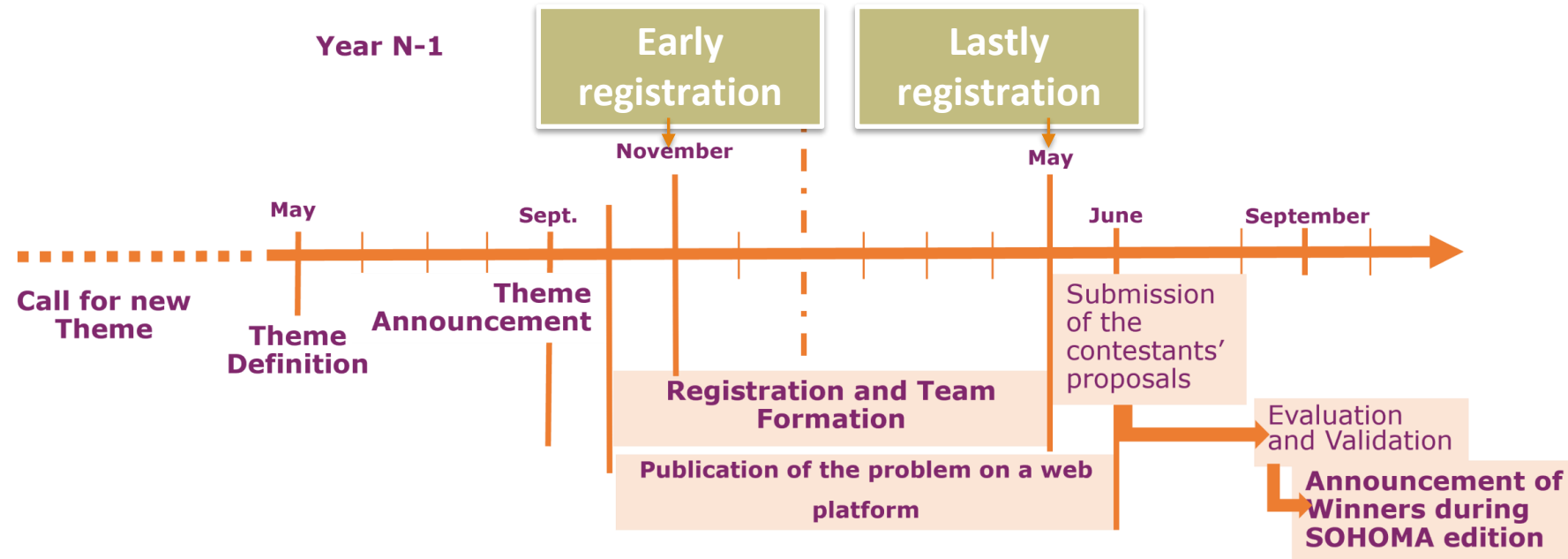
- Different problem size (Small, medium and large instances) for each category
- Report the time taken by the proposed approaches to calculate the solution.

2. Offline category:

- Inspired by the same real industrial case study
- Focuses on developing predictive approaches to find optimal solutions
 - Minimal worker intervention and maximal robot utilization.
 - Serve as benchmarks to compare with the results of the Online Category.
- **Prior knowledge** of the list of input dates and destinations of boxes in buffer A.
 - A list of tuples that represents the arrival time and destination of a box in buffer A
- The proposed approaches are developed for given scenarios of data (days/hours)
- The application will include several other scenarios with similar characteristics but varying problem sizes and random changes in the data.

IMIC26: Website & Registration

Web page : <https://github.com/GIS-S-mart/Benchmark-9-IMIC>



Award: free registration to next SOHOMA edition!

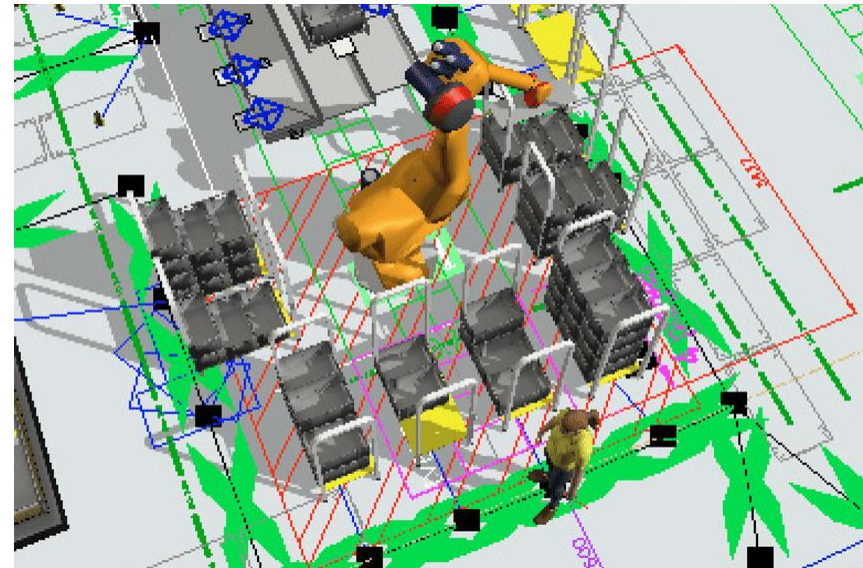
Réunion 11/02/2026 à 12h30

- Rappel motivations du concours IMIC
- Rappel problème Flexibac+
- liens utiles :
 - <https://github.com/GIS-S-mart/Benchmark-9-IMIC>
 - [IMIC'26: registration – Remplir le formulaire](#)

Où en êtes-vous ?

Quelles approches utilisez-vous ?

Avez-vous des questions ?



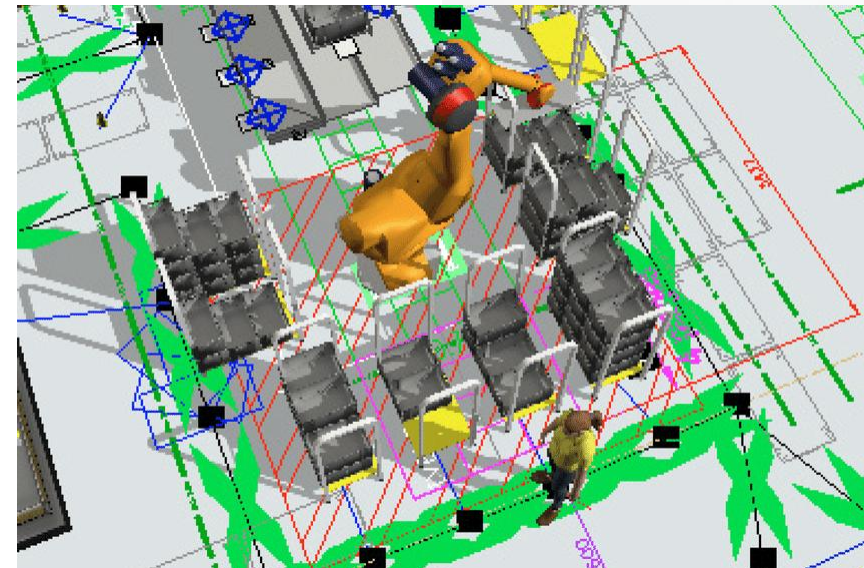
Réunion 16/03/2026 à 12h30

- Rappel motivations du concours IMIC
- Rappel problème Flexibac+
- liens utiles :
 - <https://github.com/GIS-S-mart/Benchmark-9-IMIC>
 - [IMIC'26: registration – Remplir le formulaire](#)

Où en êtes-vous ?

Quelles approches utilisez-vous ?

Avez-vous des questions ?



Réunion 05/05/2026 à 12h30

Rappel motivations du concours IMIC + problème Flexibac+

Soumission des résultats pour le 26 juin :

- Une contribution au format SOHOMA,
- Les 2 fichiers générés .dat

Où en êtes-vous ?
Quelles approches utilisez-vous ?
Avez-vous des questions ?



Annonce aux vainqueurs fin juillet

Organisation venue des vainqueurs à SOHOMA à Cambridge les 22 et 23 septembre 2026

IMIC27 :

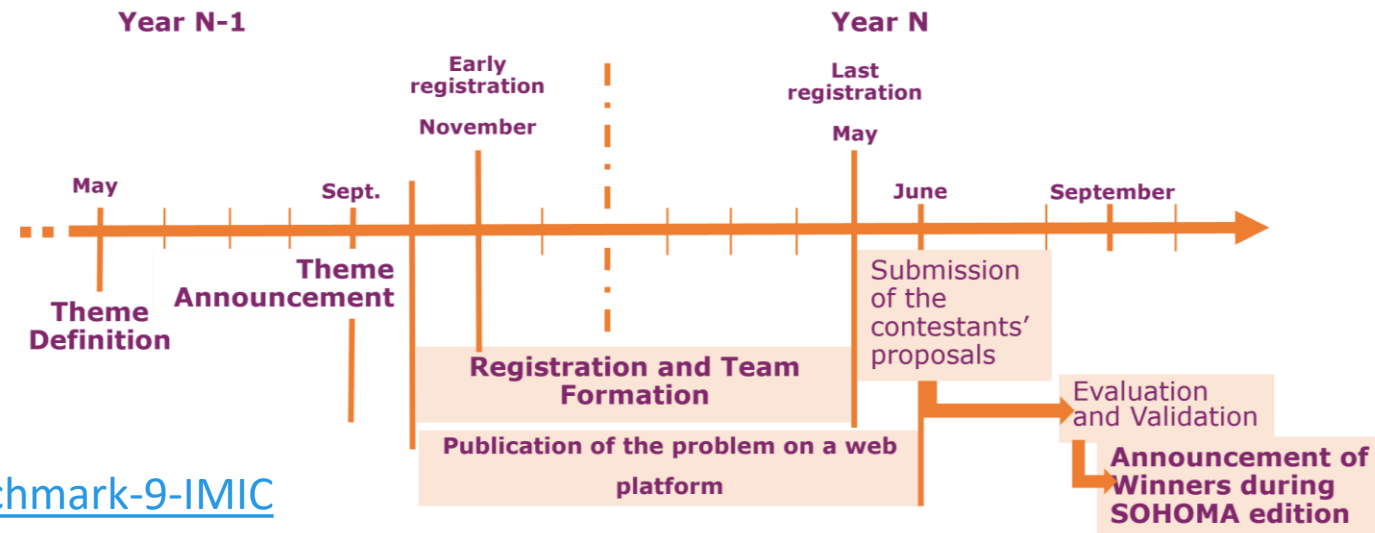
Annonce du nouveau problème à la SAGIP

Prochaine (et dernière ?) réunion : le 2 juin midi

- Liens utiles :

<https://github.com/GIS-S-mart/Benchmark-9-IMIC>

[IMIC'26: registration – Remplir le formulaire](#)



Réunion 02/06/2026 à 12h30

Rappel motivations du concours IMIC + problème Flexibac+

Soumission des résultats pour le 26 juin :

- Une contribution au format SOHOMA,
- Les 2 fichiers générés .dat

Où en êtes-vous ?
Quelles approches utilisez-vous ?
Avez-vous des questions ?

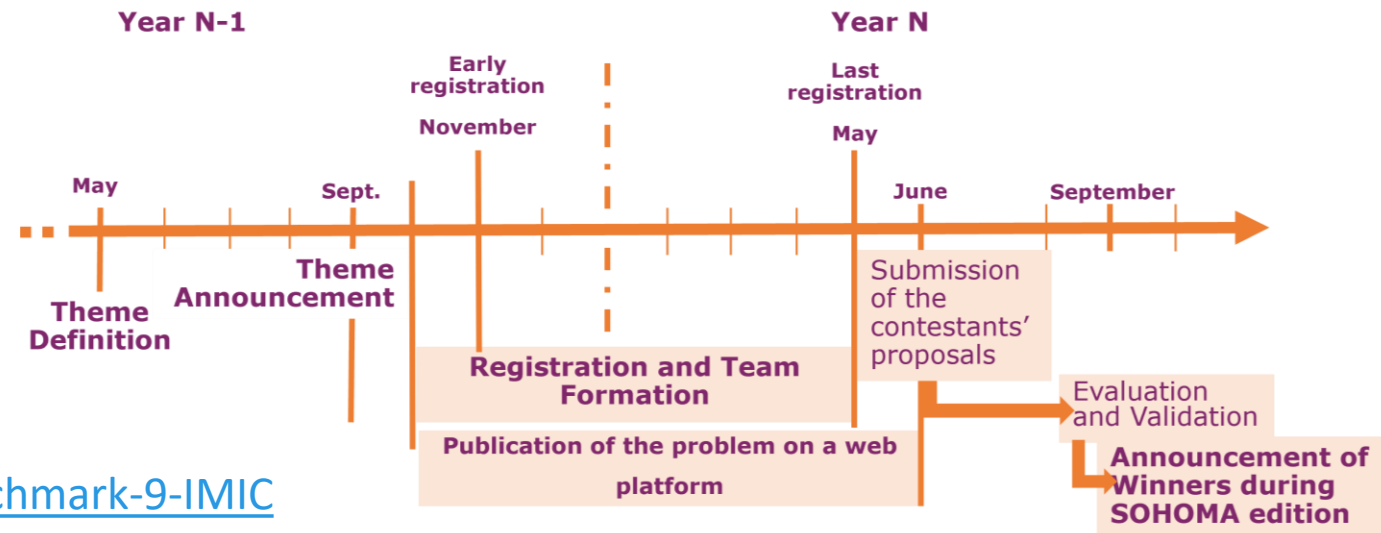


Annonce aux vainqueurs fin juillet

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IMIC27 :

Annonce du nouveau problème à la SAGIP



- Liens utiles :

<https://github.com/GIS-S-mart/Benchmark-9-IMIC>

[IMIC'26: registration – Remplir le formulaire](#)

Quelques précisions / corrections

1/ Modification des instances pour intégrer un temps de simulation de 20000s

2/ Temps de déplacement du robot = 6s!

3/ Le robot peut continuer à fonctionner lors d'un changement de chariot

1/ Modification of the timers to include a simulation time of 20,000 seconds

2/ Robot movement time = 6 seconds!

3/ The robot can continue to operate during a trolley change

Dans les papiers : mettre tous les résultats en annexes